

Weijie Chen 陈维杰

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🏠 Shenzhen, Guangdong (Permanent); Wuhan, Hubei (Current)



Education

- 2024 – now 📖 **Ph.D. Candidate, School of Cyber Science and Engineering (CSE), Huazhong Univ. of Sci. & Tech (HUST).**
Major: *Cyber Security*. Area: *System Security*. Supervisor: *Weizhong Qiang*.
- 2021 – 2024 📖 **M.Eng. (Exam Exemption), CSE, HUST.**
Major: *Cyber Security*. Supervisor: *Weizhong Qiang*. GPA: 3.49/4.0.
- 2017 – 2021 📖 **B.Eng., CSE, HUST.**
Previously the School of Computer Science. Major: *Information Security*. GPA: 3.74/4.0.

About Me

- **My research interests involve system/architectural security, micro-architectural side-channel/hardware vulnerability exploitation & mitigation, information privacy, and confidential computing. My future research plan also involves in the micro-architectural security of heterogeneous systems in ML/graphics applications.**
- Strong aptitude for independent thinking, teamwork, and communication, together with solid grasp in both English academic reading/writing and oral expression.
- Previously worked in a national/provincial key laboratory for 2+ years, gaining rich research experience as well as project application and management skills.
- My motto is 'Sic Parvis Magna' (greatness comes from tiny deeds).

Skills

- English 📖 GRE(2020): Reasoning 320/340 + Writing 4.0/6.0; TOEFL: 100;
National Qualification for Translator and Interpreter (Level-3 Written Translation, 2024)
- Coding 📖 Mainly C/C++/assembly, also skilled in Python, shell scripting and using Linux systems.
- Misc. 📖 Well-versed in the RISC-V architecture and its specifications; familiar with LLVM and its pass-based compiler customization.

Research Publications

- 1 **W. Chen**, S. Tang, Y. Tang, X. Luo, Y. Zhang, and W. Qiang, under review in the 58th IEEE/ACM International Symposium on Microarchitecture (MICRO'25, CCF-A, Top-tier), 2025.
- 2 R. Han, **W. Chen**, Y. Tang, and W. Qiang, planned submission to a CCF-B conference, 2025.
- 3 **W. Chen**, Y. Zhao, Y. Zhang, W. Qiang, D. Zou, and H. Jin, "ReminISCence: Trusted Monitoring Against Privileged Preemption Side-channel Attacks," in *Proceedings of the 29th European Symposium on Research in Computer Security (ESORICS'24, CCF-B)*, Bydgoszcz, Poland, 2024.
- 4 Y. Zhao, W. Xue, **W. Chen**, W. Qiang, D. Zou, and H. Jin, "Owl: Differential-based Side-Channel Leakage Detection for CUDA Applications," in *Proceedings of the 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN'24, CCF-B)*, Brisbane, Australia, 2024.

- 5 D. Zou, P. Zhang, W. Liu, **W. Chen**, and Y. Lu, “Universal patching method for side-channel vulnerabilities based on atomic obfuscation,” *Chinese Journal of Network and Information Security*, vol. 8(02), pp. 100–111, 2022.  DOI: 10.11959/j.issn.2096-109x.2022014.

Recent Project Experience

- 2023  **Research Intern at RITAS, Southern University of Science and Technology.** Conducted a research regarding micro-architectural side-channel monitoring on RISC-V, **advised by Prof. Yinqian Zhang** at the Teecert Lab; the paper has been **accepted by ESORICS’24**.
- 2023 – now  **Key Contributor of Research Grant No. 62272181 of the National Natural Science Foundations of China.** (*Side Channel Detection and Defense for the Trusted Execution Environment of Heterogeneous Computing Systems*).
- 2024 – now  **Key Contributor of an Internal Project related to Rowhammer Attacks.** As the principal investigator, one major work has led to the submission towards **MICRO’25 (computer architecture top-tier conference)**, collaborating with **Prof. Xiapu Luo** (Hong Kong PolyU) and **Prof. Yinqian Zhang** (SUSTech).

Miscellaneous

Awards and Achievements

- 2024  **3rd-class Scholarship for Doctoral Students, HUST.**
- 2023  **1st-class Scholarship for Postgraduates, HUST.**
- 2022  **2nd-class Scholarship for Postgraduates, HUST.**
- 2021  **1st-class Scholarship for Postgraduates, HUST.**
-  **Outstanding Undergraduate, HUST.**
- 2019  **Scholarship for Academic Progress, HUST.**
-  **2nd-class Mingde Scholarship, HUST.**

Student Work

- 2018 – 2019  **Vice President.** Publicity Department of the School of Computer Science Student Union.
- 2020 – 2021  **President.** Academic & Career Development Department of the School of Cyber Science and Engineering Student Union.
- 2024 – now  **Class Monitor** in charge of administrative affairs in my current Ph.D. cohort.

Hobbies

Beyond my area of expertise, I love seeking multi-disciplinary knowledge from diverse fields, such as biology and chemistry, in which I have received two provincial first prizes and a national third prize during high school. Also, I am a huge enthusiast of music and its theories with a good proficiency in piano. As for sports, I like swimming and have a regular fitness routine.